British Standard

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Nga Pham

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Road marking materials — Road marking performance for road users

The European Standard EN 1436 : 1997 has the status of a British Standard

ICS 93.080.20

National foreword

This British Standard is the English language version of EN 1436 : 1997 published by the European Committee for Standardization (CEN).

The UK participation in its preparation was entrusted by Technical Committee B/509, Road equipment, to Subcommittee B/509/2, Horizontal road markings and road studs, which has the responsibility to:

– aid enquirers to understand the text;

– present to the responsible European committee any enquiries on the interpretation, or proposals for change, and keep the UK interests informed;

– monitor related international and European developments and promulgate them in the UK.

A list of organizations represented on this subcommittee can be obtained on request to its secretary.

Cross-references

The British Standards which implement international or European publications referred to in this document may be found in the BSI Standards Catalogue under the section entitled 'International Standards Correspondence Index', or by using the 'Find' facility of the BSI Standards Electronic Catalogue.

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Summary of pages

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English version

Road marking materials — Road marking performance for road users

Produits de marquage routier — Performances des marques appliquées sur la route

Straßenmarkierungsmaterialien — Anforderungen an Markierungen auf Straßen

This European Standard was approved by CEN on 1997-06-20. CEN members are bound to comply with the CEN/CENELEC Internal Regulations which stipulate the conditions for giving this European Standard the status of a national standard without any alteration.

Up-to-date lists and bibliographical references concerning such national standards may be obtained on application to the Central Secretariat or to any CEN member.

This European Standard exists in three official versions (English, French, German). A version in any other language made by translation under the responsibility of a CEN member into its own language and notified to the Central Secretariat has the same status as the official versions.

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CEN

European Committee for Standardization
Comité Européen de Normalisation
Europäisches Komitee für Normung

Central Secretariat: rue de Stassart 36, B-1050 Brussels

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Foreword

This European Standard has been prepared by Technical Committee CEN/TC 226, Road equipment, the secretariat of which is held by AFNOR.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by February 1998, and conflicting national standards shall be withdrawn at the latest by February 1998.

According to the CEN/CENELEC Internal Regulations, the national standards organizations of the following countries are bound to implement this European Standard: Austria, Belgium, Czech Republic, Denmark, Finland, France, Germany, Greece, Iceland, Ireland, Italy, Luxembourg, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland and the United Kingdom.

Contents

Introduction

Road markings together with road studs form the means for horizontal signalization.

Road markings include longitudinal markings, arrows, transverse markings, text and symbols on the surface of the highway etc.

Road markings can be provided by the application of paint, thermoplastics materials, cold hardening materials, pre-formed lines and symbols or by other means.

Most road markings are white or yellow, but in special cases other colours are used.

Road markings are either permanent or temporary. The functional life of temporary road markings is limited by the duration of the road works. For permanent road markings it is best for reasons of safety to have a functional life that is as long as possible.

Road markings can be applied with or without the addition of glass beads. With glass beads the retroreflection of the marking is achieved when the marking is illuminated by vehicle headlamps.

The retroreflection of a marking, in wet or rainy conditions, can also be enhanced by special properties. The properties can be produced by surface texture (as with profiled markings), large glass beads or other means. In the case of surface texture, the passage of wheels can produce acoustic or vibration effects.

1 Scope

This standard specifies the performance for road users of white and yellow road markings, as expressed by their reflection in daylight and under road lighting, retroreflection in vehicle headlamp illumination, colour and skid resistance.

2 Normative references

This European Standard incorporates by dated or undated reference, provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this European Standard only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies.

ISO 48 : 1994	*Testing of elastomers — Determination of indentation hardness of soft rubber (IRHD)*
ISO 4662 : 1986	*Rubber — Determination of rebound resilience of vulcanizates*
ISO/CIE 10526 : 1991	*Colorimetric illuminants*
CIE 17.4	*International lighting vocabulary 1986*

3 Definitions, symbols and abbreviations

For the purposes of this standard, the following definitions apply, together with the definitions for the CIE 2° standard observer in CIE 17.4.

3.1 luminance coefficient under diffuse illumination (of a field of a road marking): Q_d ($mcd{\cdot}m^{-2}\cdot lx^{-1}$)

Quotient of the luminance of the field of the road marking in the given direction by the illuminance on the field.

3.2 coefficient of retroreflected luminance (of a field of a road marking): R_L ($mcd{\cdot}m^{-2}\cdot lx^{-1}$)

Quotient of the luminance L of the field of the road marking in the direction of observation by the illuminance $E_\perp$ at the field perpendicular to the direction of the incident light.

3.3 skid resistance tester value (of a road marking)

Skid resistance quality of a wet road surface measured by the friction at a low speed of a rubber slider upon this surface. The abbreviation SRT applies.

3.4 functional life (of a road marking)

Period during which the road marking fulfils all the requirements initially specified by the responsible road authority.

4 Requirements

4.1 General

The requirements specified aim primarily at the performance of road markings during their functional life. The requirements are expressed by several parameters representing different aspects of the performance of road markings and, for some of these, in terms of classes of increasing performance.

NOTE 1. The length of the functional life depends on whether the road marking is of short or long durability, on whether or not the road marking is run on by traffic (e.g. symbols on the carriageway compared to continuous edgelines), on the traffic density, on the roughness of the road surface and on matters relating to local conditions such as the use of studded tyres in some countries.

NOTE 2. The classes enable different priorities to be given to the different aspects of performance of road markings depending on particular circumstances. Classes of high performance cannot always be achieved for two or more of the parameters simultaneously.

4.2 Reflection in daylight or under road lighting

For the measurement of reflection in daylight or under road lighting, the luminance coefficient in diffuse illumination Q_d is used. It shall be measured in accordance with annex A and is expressed in $mcd{\cdot}m^{-2}\cdot lx^{-1}$.

Road markings in the dry condition shall conform to table 1.

NOTE. The luminance coefficient under diffuse illumination represents the brightness of a road marking as seen by drivers of motorized vehicles in typical or average daylight or under road lighting conditions.

Table 1. Classes of Q_d for dry road markings

Road marking colour	Road surface type	Class	Minimum luminance coefficient in diffuse illumination, Q_d $mcd \cdot m^{-2} \cdot lx^{-1}$
White	Asphaltic	Q0	No requirement
		Q2	$Q_d \geq 100$
		Q3	$Q_d \geq 130$
	Cement concrete	Q0	No requirement
		Q3	$Q_d \geq 130$
		Q4	$Q_d \geq 160$
Yellow		Q0	No requirement
		Q1	$Q_d \geq 80$
		Q2	$Q_d \geq 100$

NOTE. The class Q0 applies when daytime visibility is achieved through the value of the luminance factor β; see **4.4**.

4.3 Retroreflection under vehicle headlamp illumination

For the measurement of retroreflection under car headlamp illumination, the coefficient of retroreflected luminance R_L is used. It shall be measured in accordance with annex B and is expressed in $mcd \cdot m^{-2} \cdot lx^{-1}$.

Road markings in the dry condition shall conform to table 2, and shall conform to table 3 during wetness and to table 4 during rain.

NOTE. The coefficient of retroreflected luminance represents the brightness of a road marking as seen by drivers of motorized vehicles under the illumination of the driver's own headlamps.

4.4 Colour

The luminance factor β shall conform to table 5 for road markings in dry conditions. The x, y chromaticity coordinates for dry road markings shall lie within the regions defined by the corner points given in table 6 and illustrated in figure 1. Measurements shall be made in accordance with annex C.

NOTE. Measured values of the luminance factor β are not always valid for all road markings; see annex C.

4.5 Skid resistance

The skid resistance value, expressed in SRT units, shall conform to table 7. The skid resistance shall be measured in accordance with annex D.

NOTE. The test method is not valid for all types of road marking; see annex D.

Table 2. Classes of R_L for dry road markings

Road marking type and colour		Class	Minimum coefficient of retroreflected luminance, R_L mcd·m^{-2}·lx^{-1}
Permanent	White	R0	No requirement
		R2[1)]	$R_L \geq 100$
		R4[1)]	$R_L \geq 200$
		R5[1)]	$R_L \geq 300$
	Yellow	R0	No requirement
		R1[1)]	$R_L \geq 80$
		R3[1)]	$R_L \geq 150$
		R4[1)]	$R_L \geq 200$
Temporary		R0	No requirement
		R3[1)]	$R_L \geq 150$
		R5[1)]	$R_L \geq 300$

[1)] In some countries these classes cannot be maintained during a limited time period of the year during which the probability of lower performance of the road markings is high due to the presence of water, dust, mud etc.

NOTE. Class R0 is intended for conditions where the visibility of road markings is achieved without retroreflection under vehicle headlamp illumination.

Table 3. Classes of R_L for road markings in conditions of wetness

Conditions of wetness	Class	Minimum coefficient of retroreflected luminance, R_L mcd·m^{-2}·lx^{-1}
As obtained 1 min after flooding the surface with water in accordance with **B.6**	RW0	No requirement
	RW1	$R_L \geq 25$
	RW2	$R_L \geq 35$
	RW3	$R_L \geq 50$

NOTE. Class RW0 is intended for situations where this type of retroreflection is not required for economic or technological reasons.

Table 4. Classes of R_L for road markings in conditions of rain

Conditions of rain	Class	Minimum coefficient of retroreflected luminance, R_L mcd·m^{-2}·lx^{-1}
As obtained after at least 5 min exposure in accordance with **B.7** during uniform rainfall of 20 mm/h	RR0	No requirement
	RR1	$R_L \geq 25$
	RR2	$R_L \geq 35$
	RR3	$R_L \geq 50$

NOTE. Class RR0 is intended for situations where this type of retroreflection is not required for economic or technological reasons.

Table 5. Classes of luminance factor β for dry road markings

Road marking colour	Road surface type	Class	Minimum luminance factor, β
White	Asphaltic	B0	No requirement
		B2[1)]	$\beta \geq 0{,}30$
		B3[1)]	$\beta \geq 0{,}40$
		B4[1)]	$\beta \geq 0{,}50$
		B5[1)]	$\beta \geq 0{,}60$
	Cement concrete	B0	No requirement
		B3[1)]	$\beta \geq 0{,}40$
		B4[1)]	$\beta \geq 0{,}50$
		B5[1)]	$\beta \geq 0{,}60$
Yellow		B0	No requirement
		B1[1)]	$\beta \geq 0{,}20$
		B2[1)]	$\beta \geq 0{,}30$
		B3[1)]	$\beta \geq 0{,}40$

[1)] In some countries these classes cannot be maintained during a limited period of the year during which the probability of lower performance of the road markings is high due to the presence of water, dust, mud etc.

NOTE. Class B0 applies when daytime visibility is achieved through the value of the luminance coefficient in diffuse illumination, Q_d.

Table 6. Corner points of chromaticity regions for white and yellow road markings

		Corner points			
		1	2	3	4
White road markings	x	0,355	0,305	0,285	0,335
	y	0,355	0,305	0,325	0,375
Yellow road markings, class Y1	x	0,443	0,545	0,465	0,389
	y	0,399	0,455	0,535	0,431
Yellow road markings, class Y2	x	0,494	0,545	0,465	0,427
	y	0,427	0,455	0,535	0,483

NOTE. Yellow road markings class Y1 and Y2 are intended for permanent and temporary road markings respectively.

Table 7. Classes of skid resistance

Class	Minimum SRT value
S0	No requirement
S1	SRT $\geq$ 45
S2	SRT $\geq$ 50
S3	SRT $\geq$ 55
S4	SRT $\geq$ 60
S5	SRT $\geq$ 65

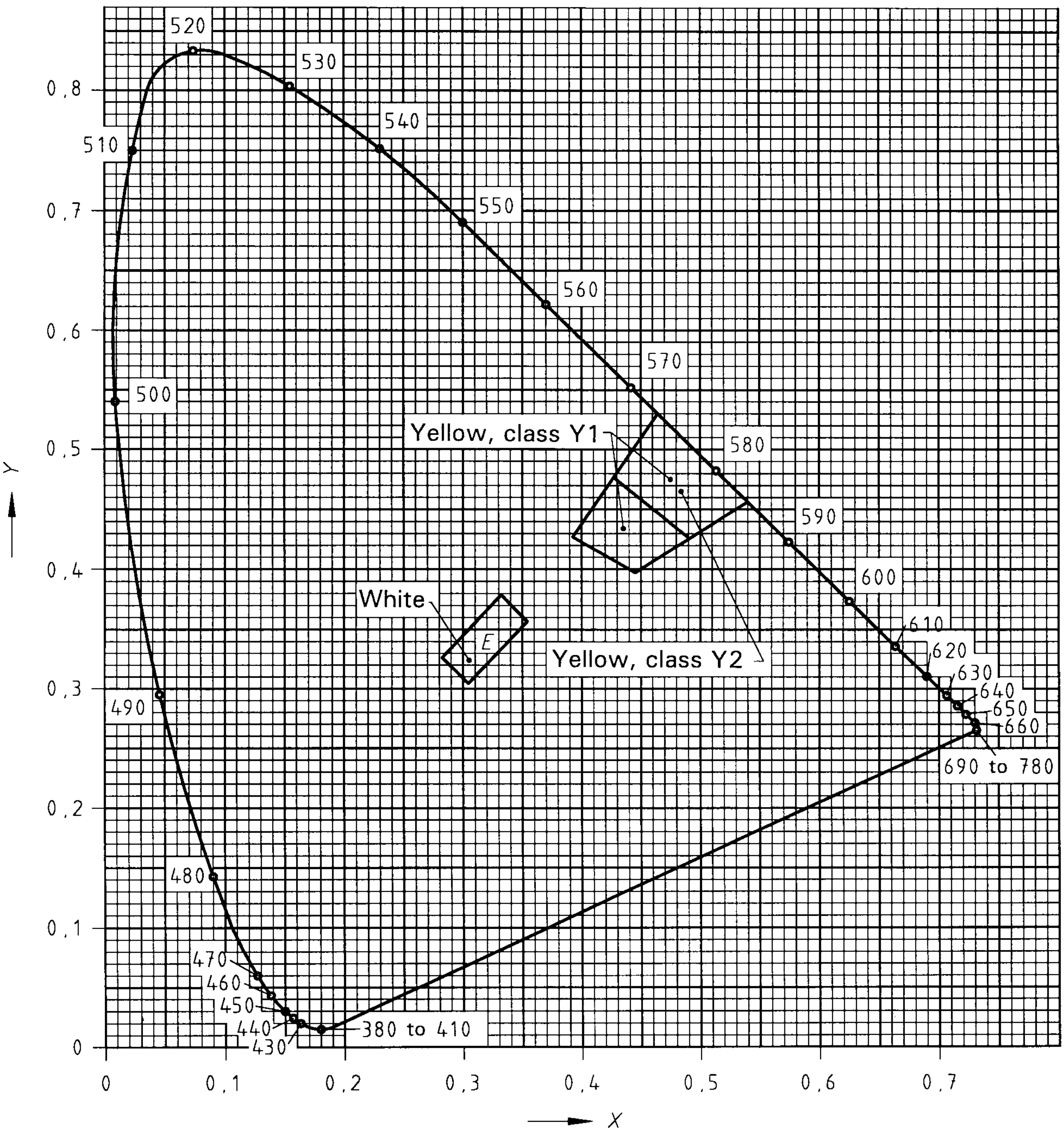

Figure 1. Chromaticity regions of white and yellow road markings in the CIE chromaticity diagram

Annex A (normative)

Measurement method for the luminance coefficient under diffuse illumination, Q_d

A.1 Standard measuring condition

The luminance coefficient under diffuse illumination, Q_d, of a field of a road marking shall be determined by:

$Q_d = L/E$ unit: mcd·m^{-2} ·lx^{-1}

where:

- L is the luminance of the field under diffuse illumination; unit: mcd·m^{-2};
- E is the illuminance on the plane of the field; unit: lx.

The luminance L shall be determined for an observation angle of 2,29° (the angle between the central measuring direction and the plane of the field) with the measuring field illuminated using standard illuminant D_{65} as defined in ISO/CIE 10526. The total angular spread of the measuring directions shall not exceed 0,33°.

The measured field of the road marking shall have an area of minimum 50 cm^2. For some profiled road markings with a considerable spacing between the profiles, the total measured field shall be long enough to include at least one such spacing. The most reliable result is obtained when the total length includes an exact multiple of the spacing. All of the measured field shall be illuminated with a uniform illuminance.

NOTE 1. The standard measuring condition is intended to simulate a visual distance of 30 m for the driver of a passenger car with an eye height of 1,2 m above the road.

NOTE 2. In real surfaces with surface texture, the measured field is elongated and often displaced. The illumination should cover all of this field.

A.2 Measurement and calibration

Measurements are in practice made by means of a $V(\lambda)$ corrected luminance meter. Diffuse illumination is obtained from an extended D_{65} light source of constant luminance, such as a photometric sphere or an illumination system with the same effect.

Calibration can be made by determining the illuminance and the luminance of the measured field. Often, a more convenient way is to measure the luminance of the illumination system by means of the luminance meter, either directly or indirectly, through a mirror. This measured value corresponds to a luminance coefficient in diffuse illumination, Q_d, of $1000\,\rho/\pi$, where ρ is the reflectance of the mirror.

NOTE. For the $V(\lambda)$ correction of luminance meters, see CIE 69.

A.3 Laboratory measurement

A.3.1 *Laboratory measurement samples*

For laboratory measurement, samples should be between 20 cm and 40 cm long, according to the measuring equipment used; but for some profiled road markings longer samples are required. Practical dimensions are 40 cm long by 20 cm wide.

A.3.2 *Method*

The sample should be backed by a plate to assist handling and should show a non-deformed surface of the road marking. The sample can either be laid directly onto the plate, or it can be taken from a road and adhered to the plate.

The diffuse illumination can be provided in a photometric sphere in which the road marking sample is mounted in a horizontal position at the centre of the sphere. A light source is mounted in the sphere in such a way that direct illumination falls only on the lower half of the sphere. By reflection and interreflection, the upper half of the sphere has a roughly uniform luminance.

A.4 Equipment for in situ measurement

For in situ measurements indirect illumination can be provided from an opening in an illuminated sphere. Other illumination may be used, provided that the luminance is constant or that it has the same effect and can be calibrated to the standard conditions.

A.5 Measurement under daylight illumination

The daylight illumination under an overcast sky, with a reasonably free view to the horizon, approximates diffuse illumination well enough to permit a measurement of the luminance coefficient in diffuse illumination.

Such measurements can be carried out by means of a luminance meter placed, for instance, in a vehicle and aimed forwards at the correct observation angle. The luminance and illuminance of road marking in front of the vehicle should be monitored simultaneously.

Annex B (normative)

Measurement method for the coefficient of retroreflected luminance, R_L

B.1 Standard measuring condition

The coefficient of retroreflected luminance, R_L, of a field of a road marking shall be determined by:

$R_L = L/E_\perp$ unit: $\text{mcd}\cdot\text{m}^{-2}\cdot\text{lx}^{-1}$

where:

- L is the luminance of the field under illumination by a single light source at a relatively small angular separation from the location from which the luminance is measured; unit: $\text{mcd}\cdot\text{m}^{-2}$;
- $E_\perp$ is the illuminance created by the light source at the field on a plane perpendicular to the direction of illumination; unit: lx.

In the standard measuring condition, the directions of measurement and illumination define a plane perpendicular to the plane of the field, the observation angle α (the angle between the central measuring direction and the plane of the field) is 2,29° and the illumination angle ε (the angle between the central illumination direction and the plane of the field) is 1,24°. The measuring field shall be illuminated by standard illuminant A as defined in ISO/CIE 10526.

The total angular spread of the measuring directions shall not exceed 0,33°. The total angular spread of the illumination directions shall not exceed 0,33° in the plane parallel to the plane of the field of the road marking, and 0,17° in the plane containing the directions of measurement and illumination.

The measured field of the road marking shall have a minimum area of 50 cm^2. For some profiled road markings, with a considerable spacing between the profiles, the total measured field shall be long enough to include at least one such spacing. The most reliable result is obtained when the total length includes an exact multiple of the spacing. All of the measured field shall be illuminated with a uniform illuminance.

NOTE 1. The standard measuring condition is intended to simulate a visual distance of 30 m for the driver of a passenger car with an eye height of 1,2 m and a headlamp mounting height of 0,65 m above the road.

NOTE 2. In real surfaces with surface texture, the measured field is elongated and often displaced. The illumination, itself being elongated and often displaced, should cover all of this field.

B.2 Measurement and calibration

Measurements are made in practice by means of a $V(\lambda)$ corrected luminance meter, while the illumination is provided by a projector lamp or a specially designed illumination system.

Calibration can be made by determining the illuminance and the luminance of the measured field. Often a more convenient method is to use a tilted reflection standard of diffuse reflection with a calibrated coefficient of retroreflected luminance, R_L. A white ceramic surface, having a calibrated coefficient of retroreflected luminance, R_L, of about 300 $\text{mcd}\cdot\text{m}^{-2}\cdot\text{lx}^{-1}$ is suitable.

NOTE 1. For the $V(\lambda)$ correction of luminance meters, see CIE 69.

NOTE 2. Care should be taken in the alignment between the road marking surface and the measuring equipment, as the measured coefficient of retroreflected luminance, R_L, is for geometrical reasons proportional to the actual value of the ratio $\sin\varepsilon : \sin\alpha$.

NOTE 3. The influence of the ratio mentioned in note 2 can be avoided when using an illuminated field contained within the measured field (the illuminated field then determines the actual measuring field and should comply with dimensions stated in **B.1**). The measured value is converted into the R_L value by multiplication with the correct value of the ratio for the standard angles:

$$\frac{\sin 1{,}24^\circ}{\sin 2{,}29^\circ} = 0{,}54$$

B.3 Laboratory measurement

Samples described in **A.3.1** are adequate for laboratory measurement of the coefficient of retroreflected luminance R_L.

B.4 Equipment for in situ measurement

For a period of 5 years after the date of publication of this European Standard, any existing measuring equipment can be used to measure the coefficient of retroreflected luminance, R_L, provided that in every individual case suitable conversion factors are used to convert measured values to values that would be obtained with the standard geometry.

B.5 Measurement under car headlamp illumination

Measurements, in accordance with **B.1**, of the coefficient of retroreflected luminance, R_L, of road markings can be made at night by using a luminance meter of suitable specifications and one of the headlamps of a passenger car at full beam, or by using a similar lamp.

NOTE 1. The measurement geometry defined in **B.1** is respected if the lamp is mounted at a height of 0,65 m above the road, the luminance meter is mounted directly above the lamp at a height of 1,2 m above the road and the measurements are made at a distance of 30 m.

NOTE 2. It is desirable that the headlamp has a luminous intensity of 100 000 cd or more so as to provide an illuminance $E_\perp$ in excess of 100 lx. The beam of the headlamp should be wide enough to allow a uniform illuminance on the measured field. A suitable measuring angle of the luminance meter is 6′, which provides an elliptical measured field of dimensions 5 cm by 130 cm. For this measuring angle, the resolution of the luminance meter should be 0,1 $\text{cd}\cdot\text{m}^{-2}$ or better.

NOTE 3. Any reflected illumination onto the calibration equipment, whether an illuminance meter or a reflection standard, should be avoided by placing screens or matt dark surfaces in front of the calibration equipment during calibration. Reflections in the road marking from luminous objects behind the road marking, such as headlamps of oncoming cars, road signs or reflecting surfaces, should also be eliminated. When measuring wet road markings it is particularly important to eliminate reflections.

B.6 Condition of wetness

The test condition is created using clean water poured from a bucket with an approximate capacity of 10 l and from a height of approximately 0,5 m above the surface. The water is poured evenly along the test surface so that the measuring field and its surrounding area is momentarily flooded by a crest of water. The coefficient of retroreflected luminance, R_L, in condition of wetness, shall be measured under the test condition 1 min after the water has been poured.

B.7 Condition of rain

Test conditions shall be created using clean water to give artificial rainfall, without mist or fog, at an average intensity of (20 ± 2) mm/h over an area that is at least twice the width of the sample under test and is of a minimum width of 0,3 m, and which is more than 25 % longer than the measuring field. The variation in rainfall between the lowest and the greatest intensity shall be not greater than the ratio 1 to 1,7.

Measurements of the coefficient of retroreflected luminance, R_L, in condition of rain shall be made after 5 min of continuous rain and while rain is falling.

NOTE 1. The intensity of the rain can be determined by measuring the volume of water collected in six flat trays within a specified period. A longitudinal row of trays can be used for the minimum width of the field of 0,3 m.

NOTE 2. Protection against wind is often required. Such protection should be open at the back in order to avoid reflections. Any mist or fog should be expelled before measurement.

NOTE 3. Measurements can be made at night in accordance with **B.5**.

NOTE 4. Measurements can also be made in the laboratory on 2 m long samples supported by rigid plates. For a realistic drainage, samples should be tilted 2 % sideways and the supporting plate should have an additional width of $(10 \pm 0{,}5)$ cm on the side tilted upwards.

Annex C (normative)

Measuring method for the luminance factor β and chromaticity coordinates x and y

C.1 Standard measuring condition

The luminance factor β and the chromaticity coordinates x and y shall be measured using standard illuminant D_{65} defined in ISO/CIE 10526. The geometry is defined at the 45°/0° situation, meaning illumination at $(45 \pm 5)°$ and measurement at $(0 \pm 10)°$. The angles are measured relative to the normal to the road marking surface. The minimum measured area of the road marking surface shall be 5 cm^2.

NOTE 1. For very rough surfaces, the area measured by the apparatus should be greater than 5 cm^2, for instance 25 cm^2.

For profiled road markings, the measured value of the luminance factor β is not always valid. The visibility in daylight or under road lighting for such road markings can only be judged by the luminance coefficient in diffuse illumination Q_d.

NOTE 2. Intermediate measuring values are the tristimuli values X, Y and Z. The stimulus Y is converted into the luminance factor β, or β is measured directly. The luminance factor is a measure of the brightness of the road marking as seen from close range. The tristimuli values are further converted into the chromaticity coordinates x and y used for specification of the chromaticity of road markings.

C.2 Measuring equipment

Measurement can be made using laboratory equipment on road marking samples or using portable equipment on road markings on the road. The equipment can either be based on direct measurement of the tristimuli X, Y and Z using filtered detectors, or on spectral measurements followed by computation of the luminance factor β and chromaticity coordinates x and y.

Annex D (normative)

Measurement method for skid resistance

D.1 Principle of the test

The test equipment consists of a swinging pendulum fitted with a rubber slider at its free end. The energy loss caused by the friction of the slider over a specified length of the road surface is measured and the result is expressed in SRT units.

For profiled road markings, a measured SRT value is not always valid. Other measures of skid resistance normally show satisfactory values for these road markings.

NOTE. The skid resistance tester simulates the performance of a vehicle with patterned tyres braking with blocked wheels at 50 km/h on a wet road. See RRL Road Note No. 27.

D.2 Description of the skid resistance tester

The skid resistance tester consists of a base with three levelling screws, a vertical column with a 508 mm pendulum of 1,5 kg mass, with a spring-loaded rubber slider mounted on the end, giving a constant force of 22,2 N on the test surface. On the column, control knobs permit the vertical movement of the suspension axis. The operator is provided with means for holding and releasing the pendulum arm so that it falls freely from a horizontal position. A 300 mm long pointer indicates the position of the pendulum throughout its forward swing and indicates the measured value on a circular scale. Two friction rings are used to bring the result to the zero of this scale for a completely free swing of the pendulum arm.

D.3 Maintenance of the rubber slider

The rubber slider is of dimensions 76,2 mm × 25,4 mm × 6,3 mm and made from rubber with the properties given in table D.1.

Table D.1 Properties of the rubber slider

Temperature °C	Resilience % Lüpke[1)]	Hardness IRHD[2)]
0	43 to 49	
10	58 to 65	
20	66 to 73	55 ± 5
30	71 to 77	
40	74 to 79	

[1)] Lüpke rebound test in accordance with ISO 4662.

[2)] International rubber hardness degree in accordance with ISO 48.

The slider can only be used for 1 year after the date indicated on its side face. One slider edge can be used for at least 100 settings (500 swings). The wear of the edge should not exceed 3,2 mm horizontal and 1,6 mm vertical, as shown on figure D.1. All new sliders should be roughened by swinging five times on a dry surface and 25 times on a wet surface (after adjusting the sliding length to between 125 mm and 127 mm).

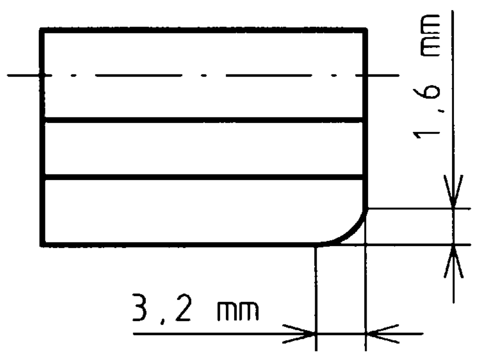

Figure D.1 Maximum wear of the rubber slider

D.4 Adjustment of the sliding length

Prior to measurement, the sliding length shall be adjusted as follows.

Ensure that the base of the apparatus is level, with the central column in front of the centre of the area under test. Raise the head so that the pendulum arm swings free of the surface and check the zero setting in this position. If necessary, adjust the setting using the friction rings so that the pointer reading is zero.

Check the sliding length (to between 125 mm and 127 mm) by gently lowering the pendulum arm until it touches the surface on one side. Place the spacer so that the outer mark on the side corresponds with the contact line between the rubber and the surface. Raise the slider off the surface by means of the lifting handle, move it without friction to the other side and then lower it gently onto the surface again. The contact line should be between the two marks on that side of the spacer. Adjust by raising or lowering the head.

When the required height is obtained, lock the head and place the pendulum in its release position.

D.5 Measuring the SRT value

The SRT value shall be measured as follows.

Wet the surface under test thoroughly and if necessary clean it with a smooth brush. Place the pendulum arm in its release position and the pointer in line with it. Release the arm and, after the maximum has been reached, catch the pendulum on its return swing with the left hand to avoid damage by striking the road surface. Read the position of the pointer. Replace the arm and the pointer in the release position.

Repeat the same measurement five times, while thoroughly wetting the contact area. If the values obtained do not differ by more than 3 units, record the mean of the five readings as the SRT value. Otherwise, repeat the test until three successive readings are constant.

Record the temperature of the water lying on the road immediately after the measurement.

D.6 Correction for the temperature

The effect of temperature on rubber resilience exerts a perceptible influence in all skidding resistance measurements; it shows itself as a fall in skidding resistance as the temperature rises. In addition, the magnitude of the variation of skidding resistance with temperature varies considerably from road to road because of the changes in road surface texture. As a rough guide, however, an average temperature correction evaluated for a range of surfaces is given in figure D.2; thus it is apparent that a correction for the effect of the temperature only becomes important for tests made below 10 °C, and then its main use is to give a more accurate assessment of the skidding resistance which the road is likely to offer to the tyres of vehicles, since they are likely to be running at temperatures rather higher than that of the slider rubber on the portable tester.

To assist in interpreting results, the temperature of the water lying on the road immediately after the test should be recorded. It should be stressed that the change in state of polish of road surfaces throughout the year is a much bigger factor determining changes in skid resistance than the change in temperature; the latter accounts for about one-quarter of the total seasonal change in skid resistance, which is primarily due to real and reversible changes in the road surface.

In order to have an idea of the influence of all variable parameters such as temperature, slider wear etc., both before and after a series of measurements, a measurement should be conducted on site with the same slider on one or more standard samples, the value of which has previously been determined in the laboratory at 20 °C.

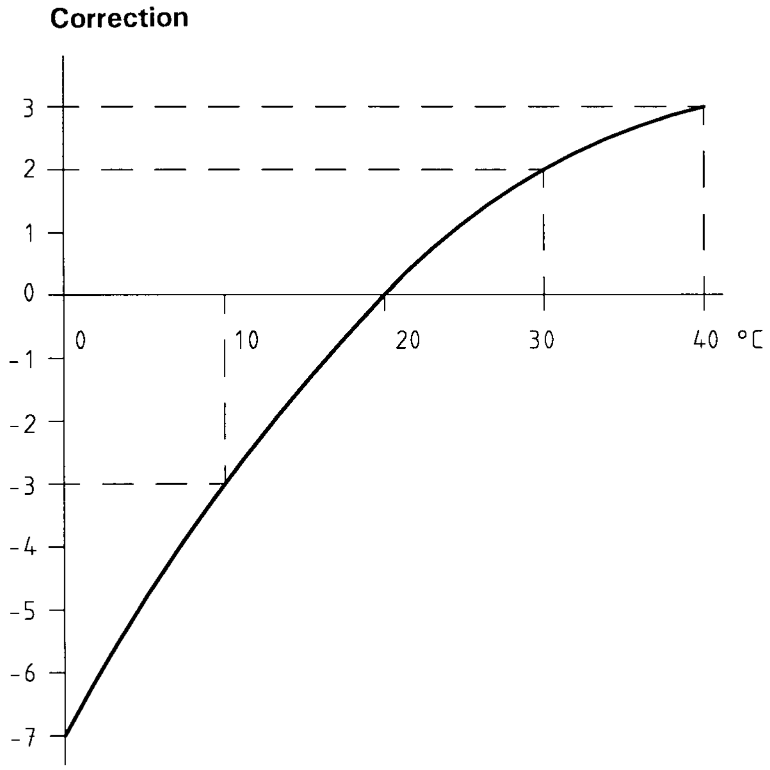

Figure D.2 Suggested temperature corrections for skid resistance values to allow for changes in resilience of the slider rubber

Annex E (informative)
Bibliography

CIE 69 : 1987 *Methods of characterising the performance of radiometers and photometers: Performance characteristics and specifications*

CIE 73 : 1988 *Visual aspects of road markings*

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